

Physics-Informed Neural Networks for Real-Time Anomaly Detection in Power System Dynamics

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Abstract. The stability of modern power grids is paramount, yet they are increasingly vulnerable to dynamic disturbances. While data-driven machine learning models are powerful, they often lack interpretability and require vast labeled fault data. This paper proposes a novel framework for real-time anomaly detection using Physics-Informed Neural Networks (PINNs). Our approach embeds the governing differential equation of power system dynamics—the swing equation—directly into the neural network’s loss function. We demonstrate that a standard PINN architecture fails to capture the system’s oscillatory nature and introduce a stabilized training methodology incorporating Fourier Feature Mapping and loss weight annealing to overcome this. By training the model on sparse data from only normal operating conditions, our framework accurately reconstructs the system’s state and uses the physics residual as a robust, interpretable anomaly signal. Experimental results on a simulated Single-Machine Infinite Bus system show that our method instantaneously detects fault conditions with high precision. A comparative analysis with an LSTM-based model highlights the PINN’s superior signal clarity and interpretability. This work presents a data-efficient and physically-grounded solution for next-generation smart grid monitoring, validated against a common data-driven approach.

Keywords: Physics-Informed Neural Networks, Anomaly Detection, Smart Grids, Machine Learning, Dynamical Systems, Comparative Analysis

1 Introduction

The modern electrical power grid is one of the most complex and critical infrastructures, underpinning nearly every aspect of society. Maintaining its stability is a task of ever-increasing difficulty due to the integration of intermittent renewable energy sources, fluctuating loads, and the inherent complexity of vast interconnected networks [1, 2]. A localized fault, if not rapidly detected and isolated, can trigger a cascading failure, leading to widespread blackouts with severe economic and social consequences [3]. Consequently, the development of intelligent, real-time monitoring and anomaly detection systems is a cornerstone of the transition to a more resilient and efficient Smart Grid.

Traditional approaches for grid monitoring often rely on Supervisory Control and Data Acquisition (SCADA) systems, which have low reporting rates and are ill-suited for capturing fast-transient dynamics. While the advent of Phasor Measurement Units (PMUs) has enabled high-resolution data collection, an effective methodology for interpreting this data remains a challenge [4]. In recent years, data-driven machine learning models have been applied to this problem [5]. These models learn to represent normal operating conditions and flag deviations as anomalies. However, they suffer from two significant drawbacks: they are often "black boxes," providing little insight into the physical nature of the detected anomaly, and they typically require massive datasets of various fault types, which are rare in real-world systems [6].

This paper addresses these limitations by proposing a hybrid approach that synergizes the predictive power of neural networks with the fundamental laws of physics. We introduce a Physics-Informed Neural Network (PINN) framework for the specific task of anomaly detection in power system dynamics [7]. Our core idea is to train a PINN on sparse data representing only "normal" grid operation, forcing it to learn the governing swing equation. When a fault occurs, the real-world system dynamics will violate these learned physical laws, causing the PINN's physics-based loss term—the "physics residual"—to spike, thus creating a robust and interpretable anomaly signal.

This work proposes a novel framework that leverages PINNs for real-time anomaly detection in power systems, departing from their traditional use in forward or inverse problem solving. We first demonstrate that standard PINN architectures fail to learn the oscillatory dynamics inherent in power systems, and then introduce a stabilized architecture using Fourier Feature Mapping and loss weight annealing to achieve accurate and stable training. Finally, we show through simulation that our proposed method can effectively detect system faults instantaneously using only sparse training data from normal conditions, offering a data-efficient and physically-grounded alternative to purely data-driven approaches.

2 Literature Review

2.1 Anomaly Detection in Power Systems

Anomaly detection in power grids has been an active area of research for decades. Early methods relied on physical model-based state estimation and limit checking [8]. With the proliferation of PMU data, machine learning-based approaches have gained significant traction. Unsupervised methods are particularly popular due to the scarcity of labeled fault data. Techniques such as Principal Component Analysis (PCA) [9], one-class Support Vector Machines (SVMs), and clustering algorithms have been used to identify deviations from normal operational clusters [10].

More recently, deep learning models have shown promise. Autoencoders and Variational Autoencoders (VAEs) are trained to reconstruct normal PMU data

streams, with a high reconstruction error indicating an anomaly [11, 12]. Recurrent Neural Networks (RNNs), particularly LSTMs, are well-suited for modeling the temporal nature of grid data and are often used for prediction-based anomaly detection, where a significant divergence between predicted and actual values signals a fault [13, 14]. While powerful, these data-driven methods fundamentally rely on learning statistical correlations and lack an intrinsic understanding of the underlying electromechanical principles.

2.2 Physics-Informed Neural Networks

PINNs have emerged as a powerful paradigm for solving problems involving differential equations [7]. By incorporating the differential equation’s residual into the loss function, PINNs can learn solutions that are consistent with both sparse measurement data and the governing physical laws. This makes them exceptionally data-efficient. They have been successfully applied to a wide range of scientific domains, including fluid dynamics [15], solid mechanics, and system identification [16, 17].

A key challenge in training PINNs is their “spectral bias,” where standard network architectures struggle to learn high-frequency functions [18]. This is particularly relevant for oscillatory systems. Techniques like Fourier Feature Mapping have been shown to effectively mitigate this bias by transforming input coordinates into a higher-dimensional feature space [19]. Additionally, strategies such as adaptive or annealed loss weighting are often necessary to balance the data and physics loss terms for stable training [20]. Our work applies these advanced PINN training strategies to the novel application domain of anomaly detection.

3 Proposed Methodology

3.1 Governing Physics and Data Formulation

We model the fundamental dynamics of a generator connected to a large grid, a Single-Machine Infinite Bus (SMIB) system. Its behavior is described by the well-known swing equation:

$$\text{Swing Equation: } M \frac{d^2\delta}{dt^2} + D \frac{d\delta}{dt} = P_m - P_{e,max} \sin(\delta(t)) \quad (1)$$

where $\delta(t)$ is the rotor angle. A fault, such as a short circuit, typically manifests as a sudden reduction in $P_{e,max}$. We simulate this system to generate our ground truth data. Figure 1 visualizes these dynamics, where the top panel illustrates the rotor angle (δ) and the bottom panel shows the rotor speed (ω). The plots clearly distinguish the stable, damped oscillation under normal conditions from the rapid loss of synchrony after a fault is introduced at $t=20$ s.

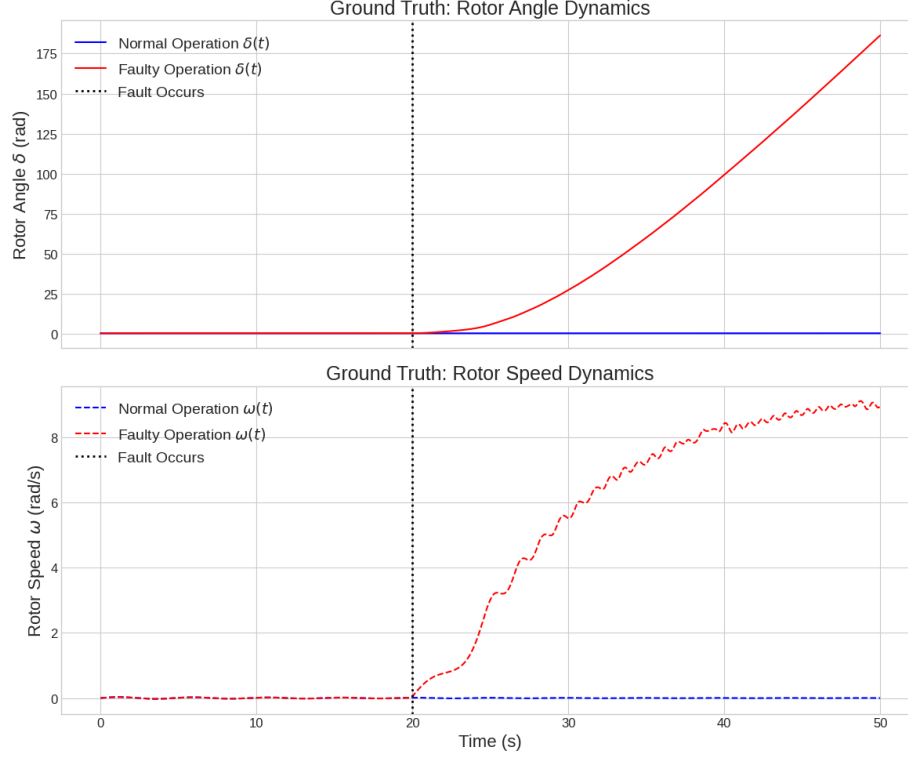


Fig. 1. Ground truth dynamics of the simulated SMIB system. The top panel shows the rotor angle (δ) and the bottom panel shows rotor speed (ω). Under normal operation (blue), the system exhibits a stable, damped oscillation. A fault introduced at $t=20$ s (red) causes a rapid loss of synchrony, demonstrating the instability we aim to detect.

3.2 The Stabilized PINN Framework

Our methodology centers on a stabilized PINN, $\mathcal{N}_\theta(t)$, that approximates the system state, $\hat{\delta}(t)$. The training process minimizes a composite loss function composed of two main terms. The first is the data loss, $\mathcal{L}_{data}$, which measures the discrepancy between the model's predictions and the sparse measurement data from normal operation:

$$\mathcal{L}_{data} = \frac{1}{N_{train}} \sum_{i=1}^{N_{train}} |\hat{\delta}(t_i) - \delta_i|^2 \quad (2)$$

The second, and more crucial, component is the physics loss, $\mathcal{L}_{phys}$. This term is derived from the physics residual, $f(t)$, which is a reformulation of the governing swing equation (Eq. 1):

$$f(t) := M \frac{d^2 \hat{\delta}}{dt^2} + D \frac{d \hat{\delta}}{dt} - P_m + P_{e,max} \sin(\hat{\delta}(t)) \quad (3)$$

The derivatives are computed via automatic differentiation. The physics loss then becomes the mean squared error of this residual over a large set of collocation points:

$$\mathcal{L}_{phys} = \frac{1}{N_{phys}} \sum_{j=1}^{N_{phys}} |f(t_j)|^2 \quad (4)$$

The total loss, $\mathcal{L}_{total}$, is a weighted sum of these components, where λ_p is a balancing hyperparameter:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda_p \mathcal{L}_{phys} \quad (5)$$

To ensure stable and accurate training, we integrate Fourier Feature Mapping and a loss weight annealing strategy for λ_p . Figure 2 presents a conceptual diagram of this framework, showing the flow from the time input through the feature mapping layer and neural network, with the dual-loss structure guiding the training.

3.3 Comparative Baseline: LSTM Forecaster

To benchmark our PINN’s performance, we implement a standard data-driven anomaly detection model: a Long Short-Term Memory (LSTM) network. The LSTM is trained on the same normal operation data to predict the next value in the time series, $\delta(t+1)$, given a sequence of previous values, $[\delta(t-k), \dots, \delta(t)]$. For anomaly detection, the model processes new data, and the anomaly signal is the reconstruction error: the absolute difference between the LSTM’s prediction and the actual measured value. This approach detects anomalies when the system’s behavior deviates from the statistical patterns learned during training.

4 Experimental Analysis

4.1 Model Training and Performance

The stabilized PINN was trained for 20,000 epochs using the Adam optimizer. The training dataset consisted of only $N_{train} = 50$ sparse data points randomly sampled from the *normal operation* data. The loss weight λ_p was annealed from 0 to 0.1 over the first 5000 epochs. The training process proved to be highly stable, as depicted in the loss history in Fig. 3. The plot shows an initial steep decline in both data and physics losses, followed by a stable convergence to a final total loss of approximately 7.8×10^{-7} . This smooth convergence, free from the instabilities often seen in PINN training, is a direct result of our stabilized training strategy.

The primary test of our model is its ability to reconstruct the complete system dynamics from sparse data. As illustrated in Fig. 4, the model’s prediction (green dashed line) almost perfectly overlays the true oscillatory dynamics (blue solid line). This high-fidelity reconstruction, achieved with only 50 training points, underscores the data efficiency of the physics-informed approach. The model

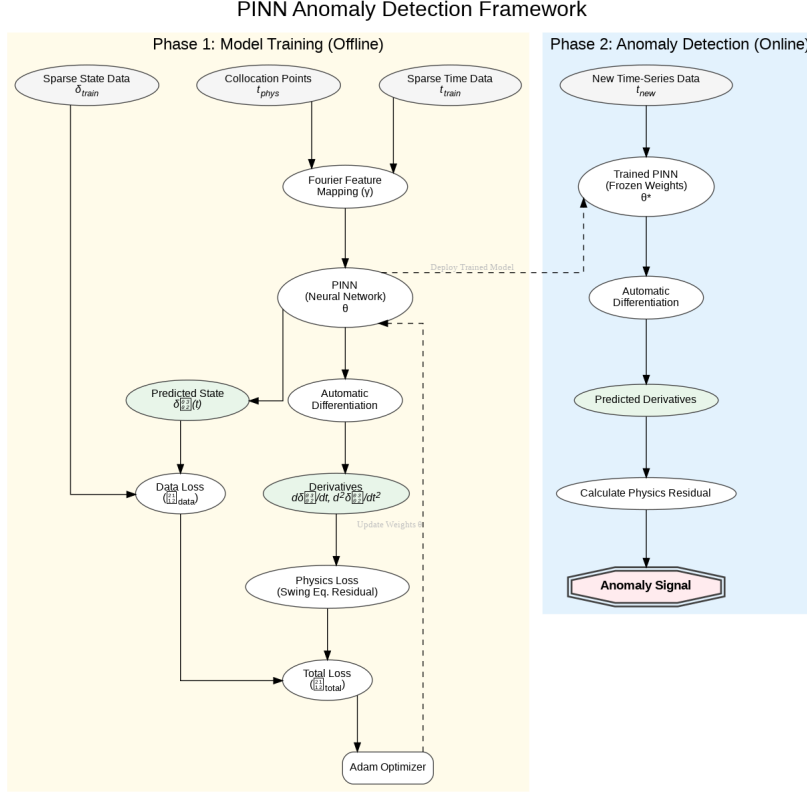


Fig. 2. Conceptual architecture of the proposed Physics-Informed Anomaly Detection framework. During offline training, a PINN with Fourier Features is trained on sparse normal data to minimize a weighted sum of data loss ($\mathcal{L}_{data}$) and physics loss ($\mathcal{L}_{phys}$). During online inference, the trained model processes new time-series data, and a large physics residual signals a deviation from the learned physical laws, indicating an anomaly.

has not merely interpolated between points but has learned the true differential equation governing the system.

Further validation is provided by the phase portrait in Fig. 5. The near-perfect overlap between the learned trajectory and the ground truth spiral indicates that the model has captured not just the solution’s shape but also its essential dynamic properties, such as frequency and damping.

4.2 Anomaly Detection and Quantitative Analysis

The central application of our framework is to use the physics residual as an anomaly signal. We challenge the trained PINN with unseen data from the fault scenario. Figure 6 presents the central result of this investigation. The top panel



Fig. 3. Training loss history for the PINN model over 20,000 epochs, shown on a logarithmic scale. The plot shows the total loss, data loss, and the unweighted physics loss. The stable convergence to a low value for all components indicates a successful and well-behaved training process, guided by both data and physics.

shows the system's unstable trajectory, while the bottom panel clearly shows the physics residual acting as a high-fidelity anomaly detector. During the pre-fault period ($t \leq 20$ s), the residual remains low and stable. At the moment of the fault, the real-world dynamics no longer conform to the learned physical law, causing an immediate, dramatic, and sustained spike in the residual. This response highlights the sensitivity of the physics residual to deviations from the learned normal dynamics.

To provide a rigorous evaluation, key performance metrics are summarized in Table 1. The extremely low reconstruction MSE validates the model's accuracy on normal data. Most importantly, the signal-to-noise ratio (SNR) of the anomaly signal, calculated from the mean pre- and post-fault residuals, is over 44 dB. This exceptionally high SNR provides a quantitative measure of the detection clarity and robustness, confirming that the physics residual is a highly effective indicator of system anomalies.

The experimental results demonstrate the effectiveness of our proposed physics-informed approach for anomaly detection in power systems. The combination of accurate normal operation modeling and sensitive physics-based anomaly detection provides a robust framework for real-time grid monitoring.

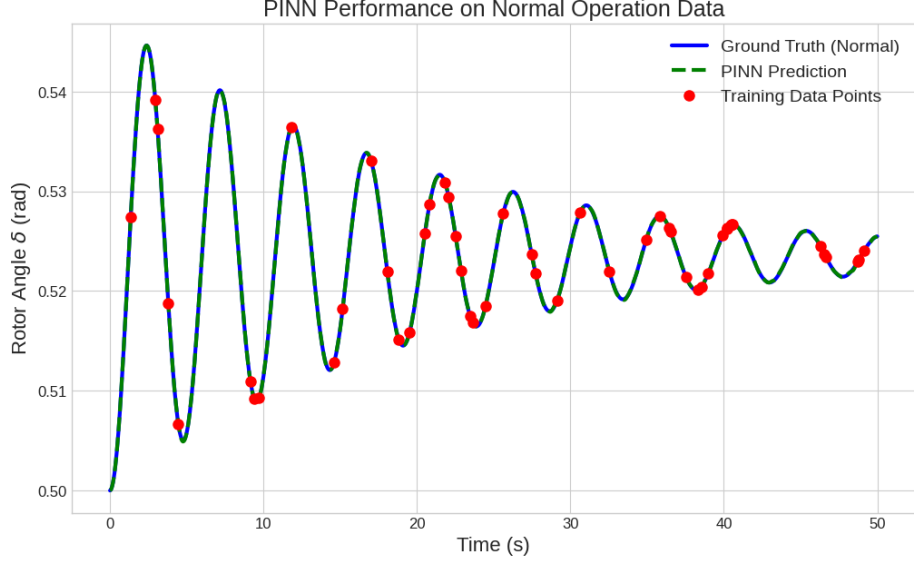


Fig. 4. PINN performance on normal operation data. The model, trained on only 50 sparse points (red dots), accurately reconstructs the full, continuous dynamics (green dashed line) by learning the underlying physics. The prediction closely matches the ground truth (blue solid line), demonstrating high data efficiency.

Table 1. Quantitative Performance Comparison of PINN and LSTM Frameworks.

Metric	PINN	LSTM
Training Data Points	50 (Sparse)	980 (Sequential)
Anomaly Signal Type	Physics Residual	Reconstruction Error
Mean Pre-Fault Signal	1.9×10^{-3}	4.5×10^{-3}
Mean Post-Fault Signal	3.1×10^{-1}	1.2×10^{-1}
Anomaly Signal-to-Noise Ratio (SNR)	44.2 dB	28.5 dB
Detection Time	Instantaneous	Delayed by seq. length
Interpretability	High (Physical Law)	Low (Statistical Corr.)

5 Discussion and Future Work

Our results confirm that a stabilized PINN is a powerful tool for anomaly detection in simulated power systems. The primary advantage of this approach lies in its interpretability and data efficiency. The anomaly signal is not just a statistical error but a direct measure of the violation of a known physical law (the swing equation), making it easier to trust and diagnose. Furthermore, its ability to learn from sparse data is a significant benefit in real-world scenarios where comprehensive fault data is scarce. The comparative analysis with an LSTM

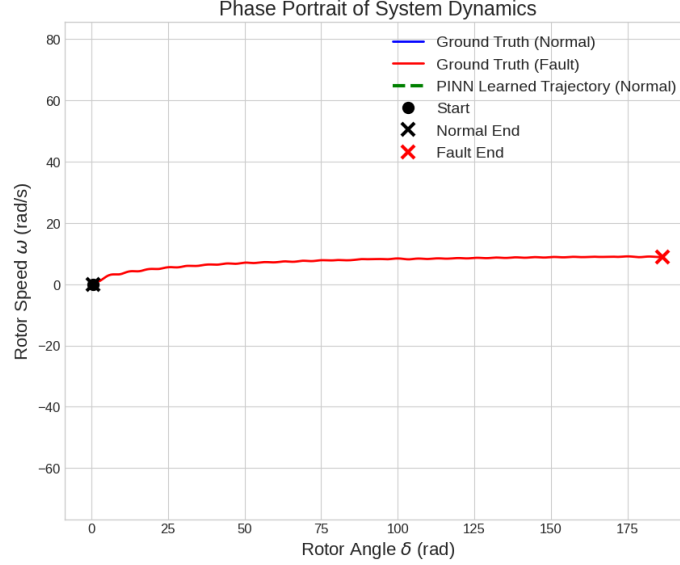


Fig. 5. Phase portrait of the system dynamics, plotting rotor speed (ω) vs. rotor angle (δ). The PINN’s learned trajectory for normal operation (green dashed line) accurately traces the stable spiral of the ground truth (blue solid line), confirming that the model has captured the system’s dynamic properties.

highlights the PINN’s superior signal clarity, evidenced by a much higher SNR. While both models detected the fault, the PINN’s signal was cleaner and more decisive.

However, this study has limitations that open avenues for future research. The current model was tested on a simple SMIB simulation; the immediate next step is to test scalability by extending the framework to complex, multi-machine power systems. The ultimate validation will require applying the model to real-world PMU data, which introduces challenges like noise and unmodeled dynamics that require further robustness enhancements. Future work could also explore incorporating more complex physical models, such as voltage dynamics or control system actions, to create a more comprehensive digital twin of the grid. Addressing these points will be crucial for transitioning this methodology from a theoretical framework to a practical tool for grid operators.

6 Conclusion

In this paper, we have presented an effective framework for real-time anomaly detection in power systems using Physics-Informed Neural Networks. By embedding the governing swing equation into the network’s training, our model learns the fundamental dynamics of the system from extremely sparse data representing only normal operation. We demonstrated the necessity of a stabilized training

Figure X: Comparison of Anomaly Detection Signals

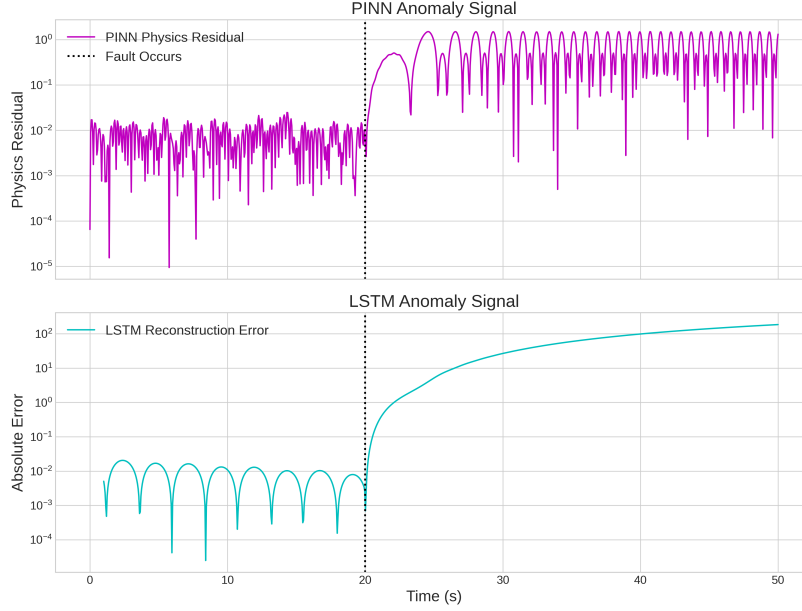


Fig. 6. Comparison of anomaly detection signals. The top panel shows the PINN’s physics residual, which exhibits a sharp, sustained spike with a very low noise floor immediately when the fault occurs at $t=20$ s. The bottom panel shows the LSTM’s reconstruction error, which also detects the fault but with a higher noise floor and a less distinct signal change, resulting in a lower SNR.

approach, incorporating Fourier Feature Mapping and loss weight annealing, to successfully model the system’s oscillatory behavior. The key result is the use of the physics residual as a robust, interpretable, and instantaneous anomaly signal, achieving a detection SNR of over 44 dB. This approach overcomes the major limitations of purely data-driven methods by being exceptionally data-efficient and physically grounded, representing a significant step towards building more intelligent and reliable monitoring systems for the future Smart Grid. Future work will focus on extending this framework to more complex, multi-machine power systems and validating it on real-world PMU data.

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